

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior backend engineer (Node.js + TypeScript) with deep migration and reliability experience — owned a **zero-downtime migration of 20+ production apps** at MTailor with idempotent, concurrent Pub/Sub-based two-way sync; ran 2M-record analytical jobs at Dr Wealth; architected Kafka + Kubernetes microservices for 4 NGO tenants at CodeBits. SQL-fluent (Postgres-ready via Firestore migration depth), AWS S3/EC2/EBS production background, GCP serverless equivalent to AWS Lambda. Strong fit for high-transaction-volume SaaS where concurrency, idempotency and performance under load matter.

CORE COMPETENCIES

Node.js + TypeScript	PostgreSQL / SQL	AWS (S3, EC2, EBS)	Serverless / Cloud Functions	Concurrency + idempotency
Data migrations at scale	Kafka + Pub/Sub	FinTech (DeFi adjacent)		

WORK EXPERIENCE

MTailor

September 2022 – Present

Software Engineer (US Remote) — reporting to CTO

- Led **zero-downtime migration of 20+ production Node.js apps** from Parse/MongoDB to Firebase/GCP — full SDLC ownership from architecture to deployment
- Built **idempotent, concurrent two-way sync between MongoDB and Firestore via Google Pub/Sub** — high-volume message processing with drift detection during live migration
- Implemented a 3D visualization feature using video overlay and ffmpeg, increasing buyer conversion
- Migrated file storage from **Amazon S3 to GCS** via a Python script with retries and dry-run mode
- Implemented Express Shipping integration, generating \$40 additional revenue per order
- Saved the company **\$5,000/month** by migrating all services off AWS

CodeBits

January 2020 – July 2021

Software Engineer / Lead (Uganda Remote)

- Led a team of 4 developers building legal-tech systems for NGOs in Uganda
- Architected microservices backend with Apache Kafka, Docker, Kubernetes, NATS Streaming, and gRPC** — multi-tenant scalable system patterns equivalent to AWS EKS + SQS/SNS
- Built a case management system for FIDA Uganda and a paralegal database for LASPNET Uganda
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC
- Developed a cross-platform mobile app for LASPNET using React Native and Expo

Mind2matter

January 2022 – July 2022

Software Engineer (US Remote) — reporting to CTO

- Built backends for **DeFi / FinTech applications using Web3 and Node.js** — wallet signing, smart-contract integration
- Delivered React UIs from Figma designs under tight timelines

Dr Wealth

August 2021 – January 2022

Software Engineer (Singapore Remote)

- Optimized a PWA delivering real-time stock prices from a Firebase backend
- Built ad-hoc jobs to query over 2 million Firestore records under SLA-bound windows — performance optimization under load
- Extended backend APIs using Firebase Cloud Functions (serverless) and Express.js on Heroku

MTailor

March 2022 – July 2022

Contractor (US Remote)

- Built a Node.js migration tool reading MongoDB and writing Firestore after complex transformation logic — schema modeling at scale
- Wrote SDK documentation for fellow engineers; trained Ops team on Firebase Dashboard

EDUCATION

B.Sc. Computer Engineering · [Makerere University, Kampala](#)

2018 – 2022

SKILLS

Proficient: Node.js, TypeScript, JavaScript, PostgreSQL/SQL, MongoDB, AWS (S3, EC2, EBS), GCP (Firestore, Pub/Sub, Cloud Functions, GCS), serverless, Python, Git, Linux, Kafka, Docker, Kubernetes

Intermediate: Go, Rust, React, React Native, gRPC, NATS, Redis (familiar), AWS EKS / SQS / SNS (transferable from K8s + Kafka)

Languages: English (native), Kiswahili